PROPOSED MARKING GUIDE

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Class	Stream
S. 1	

553/1
BIOLOGY
(Theory)
Paper 1
2 hours

Educan

EDUCAN SCHOOL

Uganda Lower Secondary Certificate of Education.
(U.L.S.C.E)

S.1 COMPETENCE BASED ASSESSMENT EXAMINATIONS
BIOLOGY
(THEORY)
Paper 1
2 Hours

INSTRUCTIONS:

- *This paper consists of* **5** questions which are ***ALL*** *Compulsory*
- *Answer* ***all*** *questions in the spaces provided.*
- *Illustrations in form of drawings should be made where necessary, with a sharp pencil.*

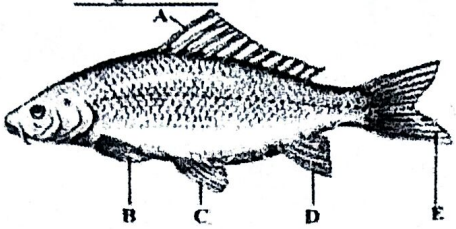
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Number	*Score*	*Teacher's comment*
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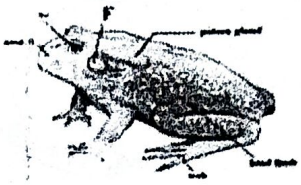
SECTION A

1. The following are living organisms that we e collected from the environment. They have been labeled as organism **X**, organism **Y** and org nism **Z**; but these are not their real taxonomic names. Study them carefully and give appr priate responses to some questions that will follow.

Organism X



rganism Y



Organism Z



a) Name the parts labelled A – F. **(03 marks)**

A. Dorsal fin ; ✓ B. Pectoral fin ; ✓

C. Pelvic fin ; ✓ D. Anal fin ; ✓

E. Caudal / Tail fin ; ✓ F. Ear drum / Tympanic membrane ✓ 03

b) The organisms shown above are classified as a imals. Outline any two observable features that may qualify them to be called animals. **(02 marks)**

Have eyes. ✓

Have head ; ✓ Have locomotory devices / fins / legs ✓ 02

c) Observe organisms X and Y. state any three di ferences in structure between the two organisms. Write your answers in the table below; **(03 marks)**

Organism X	*Organism Y*
Has fins ✓	Lacks fins. ✓
Have scales ✓	lacks scales. ✓
Lacks ear drum / tympanum ✓	Has ear drum / tympanum ✓

d) Organism X spends the whole of its life in wa r. However, this organism has some features that enable it to live in water. Identify any **two** of suc features from the diagram. **(02 marks)**

Has fins for swimming in water. ✓

Has stream-lined body ; easily swim in water ✓

Has gills ; for breathing in water. ✓

2. Mr. Mukundane, a typical farmer from Kabaale district has bought a new piece of land to expand his farm. He has however made some strange observations in the new area that when it rains, the soil becomes dry just shortly after it has rained. The crops he had planted have even started to dry up due to this very high drainage of the soil.

(a) With a reason, suggest which soil type could be making up the highest composition of the soil in the area. **(01 mark)**

Sand soil; because it has highest drainage.

(b) State any other two properties of this soil type other than high drainage. **(02 marks)**

Large soil particles; Large air spaces; Low capillarity; Less organic matter; Light brown colour;

Acc. Any correct property of sand soil

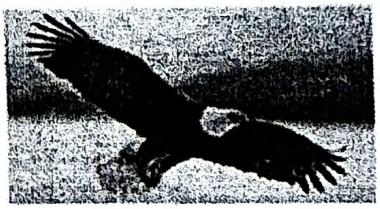
(c) What other challenges are the crops most likely to face apart from lack of enough water? **(02 marks)**

Malnourishment due to few/less nutrients in the soil.
Chlorosis/Yellowing of leaves

(d) How can Mr. Mukundane improve the drainage of the soil in his garden? **(03 marks)**

Adding organic matter to the soil;
Adding fertilisers; like NPK;

3. Below are some of the photos of some organisms that were sighted around a national game park in Uganda. Study them carefully and answer the questions that follow;



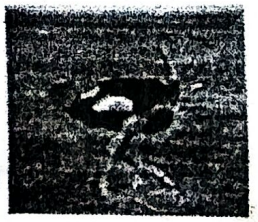
Eagle



Crested Crane



Cork



Ostrich

(a) Classify the organisms according to the following categories; (03 marks)

(i) Kingdom.: Animalia;

(ii) Phylum: Chordata/Vertebrata;

(iii) Class.:

Aves; ✓

(b) Using only observable features in the photo s, state any **two** reasons for the class in (iii) above of the organisms. **(02 marks)**

Walk on two legs; ✓
Have feathers on body; ✓
Have beaks / mouth modified into a beak; ✓

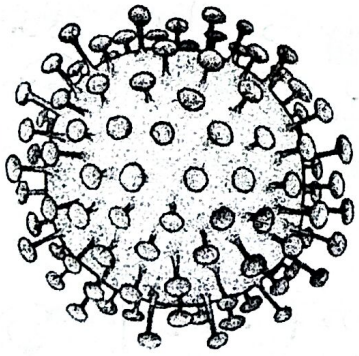
(c) State any **two** adaptations of the organisms to their way of life. **(02 marks)**

Have wings for flight; ✓
Feathers all over their body; to keep them warm ✓
Have eyes for seeing and capturing food. ✓

(d) Of what national importance is the crested crane to Uganda? **(03 marks)**

- Crested crane is the national emblem of Uganda ✓
- Tourist attraction; brings foreign currency into the County ✓
- For decorative purpose due to its nice colours. ✓

4. In 2019, a global pandemic broke up and affected almost every country in the world. There were lots of deaths across the world, nd prolonged lock downs that even caused closure of schools in Uganda. The disease was caused by a strange virus that scientists later isolated as having the structure as showed below;



(a) Write down the name of the virus shown in the diagram. **(01 mark)**

Corona Virus; ✓

(b) Name the disease caused by the above pathogen. **(01 mark)**

COVID-19; ✓

(c) State **two** ways in which the above pathogen can be spread from one individual to another. **(02 marks)**

- Inhaling droplets from a sick person; ✓
- Touching infected surfaces and then touch your nose; ✓
- Kissing an infected person; ✓

(a) Give **three** signs and symptoms of the disease caused by the above virus. **(03 marks)**

Severe fever;
Dry cough;
Loss of taste;
General body weakness;

(b) How can the spread of the above pathogen be controlled? Give three ways. **(03 marks)**

- Clean and disinfected frequently touched surfaces;
- Avoid being in crowds;
- Wearing a face mask whenever in public;
- Wash hands with soap and clean water regularly;

SECTION B

Attempt only one question from this section

5. Wepkhulu's dad felt pain in his body and he suspected that he had malaria. He visited a clinic for a medical check-up. The doctor requested a laboratory technician to obtain a blood sample from him. He found out that a number of red blood cells had been affected by the malaria parasite. On his way back, Wepkhulu's dad met the village chairperson. He narrated to him what had transpired in the clinic. The chairperson told him that other people in the village were experiencing similar symptoms. The chairperson admitted that he did not know about cells.



Task

Using your knowledge and understanding of cells, help Wepkhulu to design a poster he can give to his dad to help the chairperson to educate the village about cells. **(10 marks)**

6. It has been noted that sugar scarcity in the country has progressively increased, and leaders are trying to find ways of curbing this problem. There has been a decision taken by the district administration to give away the government land next to your school to Mr. Sanje, an Indian Investor dealing in sugar production industry, to build a sugar factory and also plant sugar cane. This land covers a total area of $5km^2$, and is the biggest natural forest in your district. The school community is however concerned about this issue





Task:

You are the president of the Nature club in your school and you have been tasked to write a letter to the district forestry officer.

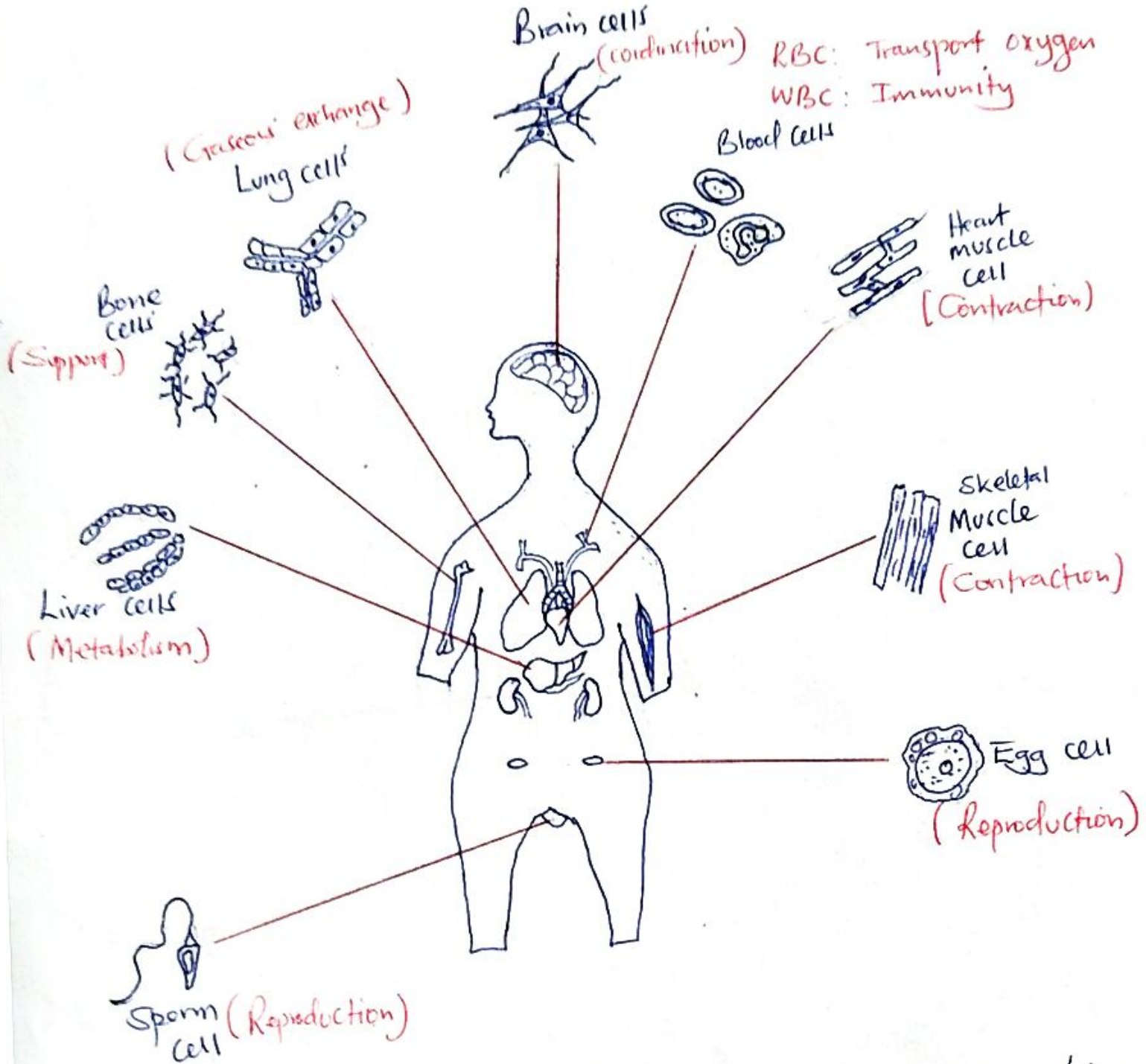
Using your knowledge of life processes, write a letter to the district forestry officer, outlining clearly why it is important that the forest be conserved. **(10 marks)**

END

"The people that finally get to the top of their dreams are those that keep pushing forward, even when the tides seem to be going against them"

QUESTION 5

CELLS CELLS CELLS



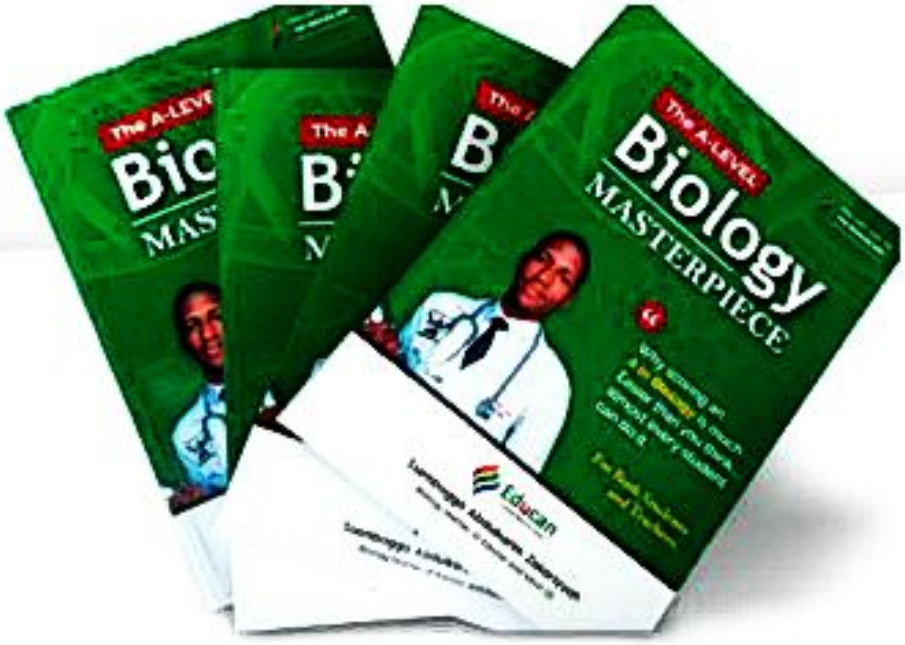
Cells are the basic units of living organisms. Just like buildings are made up of bricks, held together with sand, all living organisms are made up of units called cells. Cells vary in size, shape and function and have different functions.

EDUCAN PUBLICATIONS

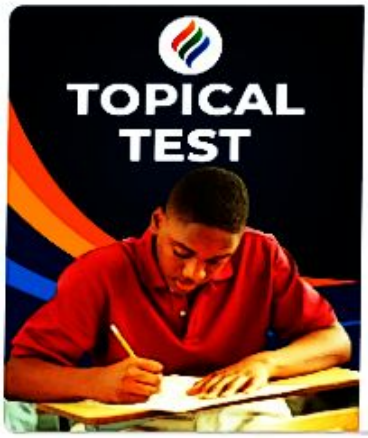
@10K



@30K

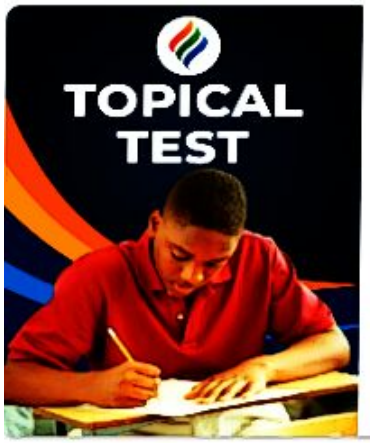


Daily Topical Tests



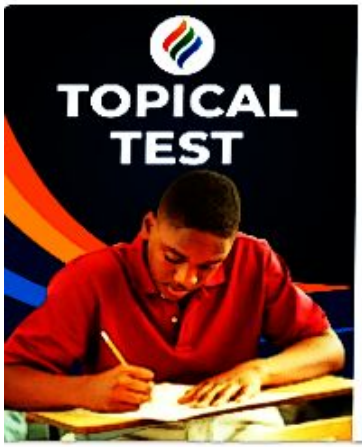
CBC BIOLOGY EXAM 002

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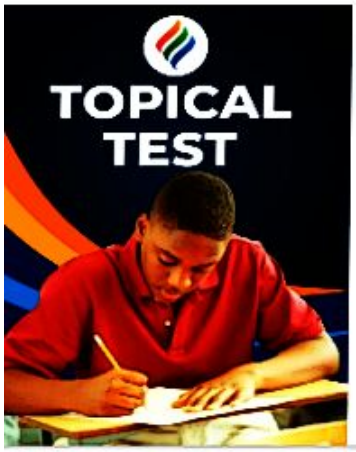
S.2 BIOLOGY CBC EXAM 001

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TEST 020

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TEST 019

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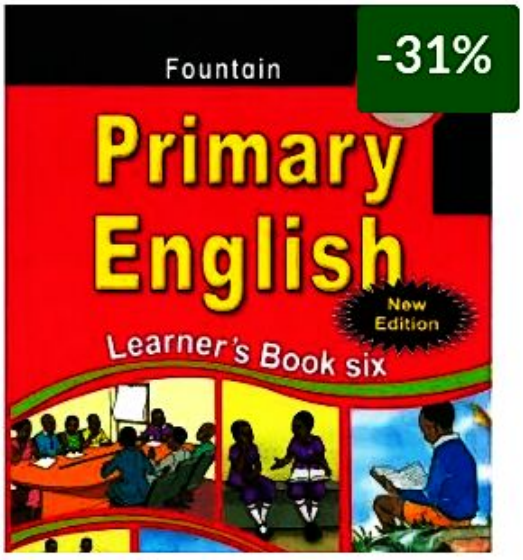
Educan Talk

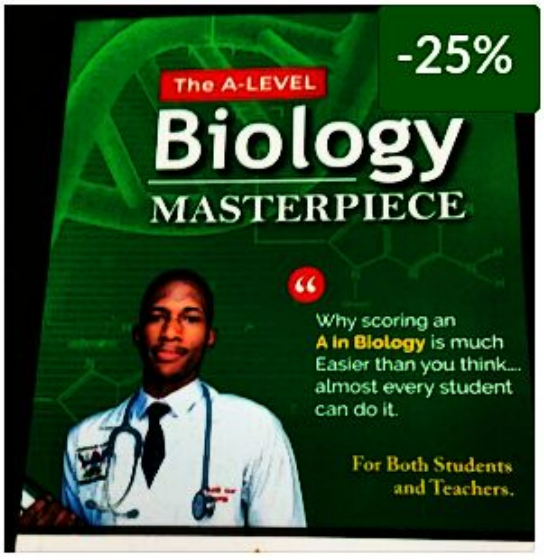


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